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Review

Genes on B chromosomes: Old questions revisited with new tools

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ABSTRACT

Background: B chromosomes are supernumerary dispensable parts of the karyotype which appear in some individuals of some populations in some species. Often, they have been considered as 'junk DNA' or genomic parasites without functional genes.
Scope of review: Due to recent advances in sequencing technologies, it became possible to investigate their DNA composition, transcriptional activity and effects on the host transcriptome profile in detail. Here, we review the most recent findings regarding the gene content of B chromosomes and their transcriptional activities and discuss these findings in the context of comparable biological phenomena, like sex chromosomes, aneuploidy and pseudogenes.
Major conclusions: Recent data suggest that B chromosomes carry transcriptionally active genic sequences which could affect the transcriptome profile of their host genome.
General Significance: These findings are gradually changing our view that B chromosomes are solely genetically inert selfish elements without any functional genes. This at one side could partly explain the deleterious effects which are associated with their presence. On the other hand it makes B chromosome a nice model for studying regulatory mechanisms of duplicated genes and their evolutionary consequences.

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1. What are B chromosomes?

B chromosomes (Bs) are supernumerary dispensable parts of the genome that are present in 15% of all eukaryotes [1]. They occur in a wide range of taxa from fungi to plants and animals including mammals [2]. Although dependent on the species, they may vary in size, structure and chromatin properties, but nevertheless they share certain characteristics which make them unique and distinguishable from other types of chromosome polymorphisms like for example aneuploidy.

Bs are dispensable and hence can be present or absent in individuals of the same population of a given species. They do not recombine with the standard set of A chromosomes (As) at meiosis and therefore have their own evolutionary pathway (reviewed in [3]). While unbalanced numbers of As like in the case of aneuploidy often cause severely distorted phenotypes [4], the presence of Bs is associated with mild or no obvious phenotypes in many species. This feature led to the conclusion that Bs are depleted of functional genes. However, an excessive number of Bs can cause phenotypic effects and may reduce the fertility of the host [2]. The maximum number of Bs tolerated by individuals varies among different species. Maize, for instance, can carry up to 34 Bs, with the number of Bs even exceeding the number of As in this case [2,5]. The harmful effects of Bs on fitness were charged to the energy cost of maintenance of Bs and their potential interference with the proper assortment of As during meiosis [2]. Besides reduction of fitness, further phenotypic effects associated with the presence of Bs have been reported. The contributions of single Bs are usually cumulative, with the severity of effects increasing with the number of Bs (reviewed in [2,6–8]). However, in cichlid fishes [9], Bs are likely to play a role in sex determination, and in the fungus *Nectria haematococca* Bs account for antibiotic resistance and pathogenicity [10]. Recently, it was shown that the presence of Bs diminishes the activity of antioxidant enzymes in calli of the plant *Acanthophyllum laxiusculum* under salinity stress conditions [11]. The question then arises if they are not necessary for normal development of their carrier, how Bs remain in a population? To counteract elimination, Bs of many species have evolved drive mechanisms which assure their transmission to the next generation at frequencies that are higher than expected according to Mendelian rules (reviewed in [12,13]). The balance between such a chromosome drive acting pre- or postmeiotically and the negative effects that the presence of Bs exerts on the fitness of their bearer, in particular when they are present in higher numbers, determines the incidence of Bs in a particular population.

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2. How does a B chromosome evolve?

The question of the origin of B chromosomes arose with the first discovery of these supernumeraries and still remains unanswered for the majority of Bs. Generally, it is assumed that Bs are derived from standard chromosomes, either from the same (intraspecies origin) or from a related (interspecies hybrid origin) species. The available experimental data support both scenarios (reviewed in [14]), suggesting that the way of their formation differs between different species and B chromosome types. Recent studies in rye [15] and in fish species *Astatotilapia latifasciata* and *Astyanax paranae* [16,17] using next generation sequencing and sophisticated bioinformatics tools (e.g. genome zipper [18] and repeat clustering analysis [19]) showed that in the analyzed systems, Bs had originated from multiple As, most likely by a pathway involving partial genome duplication and chromosome rearrangements. Similarly, analysis of evolutionarily conserved A chromosome segments in wild canid species identified several regions of domestic dog sequences that share sequence similarity with canid B chromosomes [20].

Detailed analysis of rye B-located high copy sequences revealed that Bs contain a similar proportion of repeats as A chromosomes, but differ substantially in their repeat composition [15,21,22]. B-specific accumulation of *Gypsy* retrotransposons and other repeated sequences has also been reported for the fish *Alburnus alburnus* [23], Maize [24,25,26], rye [27] and the fungus *N. haematococca* [10]. The accumulation of active mobile elements on Bs might have its cause in a relaxed selection pressure, as any functional genes on Bs should be redundant to their counterparts on the A chromosomes. Reduced crossing-over might further facilitate retroelement accumulation, as has been proposed for sex Y chromosomes [28]. To explain a B-specific absence of a mobile element, a possible scenario might involve transposition of elements predominantly in plants without Bs [21]. The element could still transpose in 0B individuals, thus allowing accumulation on As exclusively. A similar behavior has been proposed for the Y chromosome depleted *Ogre*-element in *Silene latifolia* [29]. In addition, Bs of rye [15] and the wild wheat *Aegilops speltoides* [30] have accumulated significantly greater amounts of chloroplast- and mitochondrion-derived sequences than the A chromosomes in these species. The higher amount of organelle-derived DNA inserts in Bs than in As and an increased mutation frequency of B-located organellar DNA are consistent with the assumption of a reduced selection in supernumerary chromosomes [31].

A multi-step model for the multi-A chromosome-derived origin of B chromosomes has been proposed for the Bs of different species. To trace the origin of the rye B we analyzed the genic B sequence reads against the virtual gene map of rye [32] to depict the positional information on the respective A chromosomes (Fig. 1, Martis and Gundlach, unpublished). The analyzed rye B chromosome shows the most extended sequence overlaps with the A chromosomes 3R and 7R, along with thousands of short genic sequence stretches apparently derived from all A chromosomes. Likely, initially, a proto-B chromosome was formed by segmental genome duplication (e.g. trisomy) and interspecies hybridization event, followed by reductive chromosome translocations, unbalanced segregation of a small translocation chromosome and subsequent sequence insertions. Alternatively, the centromeric "by-product" of a Robertsonian translocation between two non-homologous acrocentric chromosomes with breakpoints close to centromeres could evolve into a proto-B. Indeed, analysis of mammalian Bs showed that Bs are more common in species with acrocentric chromosomes [33,34]. After proto-B formation recombination with donor As became restricted, probably due to multiple rearrangements and illegitimate recombination involving different As, which precluded extended pairing with the formerly homologous A chromosome regions. This restriction of recombination may be considered as the starting point for the independent evolution of Bs. At the same time, a drive or a fitness benefit of the nascent B is a prerequisite for an involving B, as simulation studies performed by Camacho and colleagues demonstrated [35]. Whether the

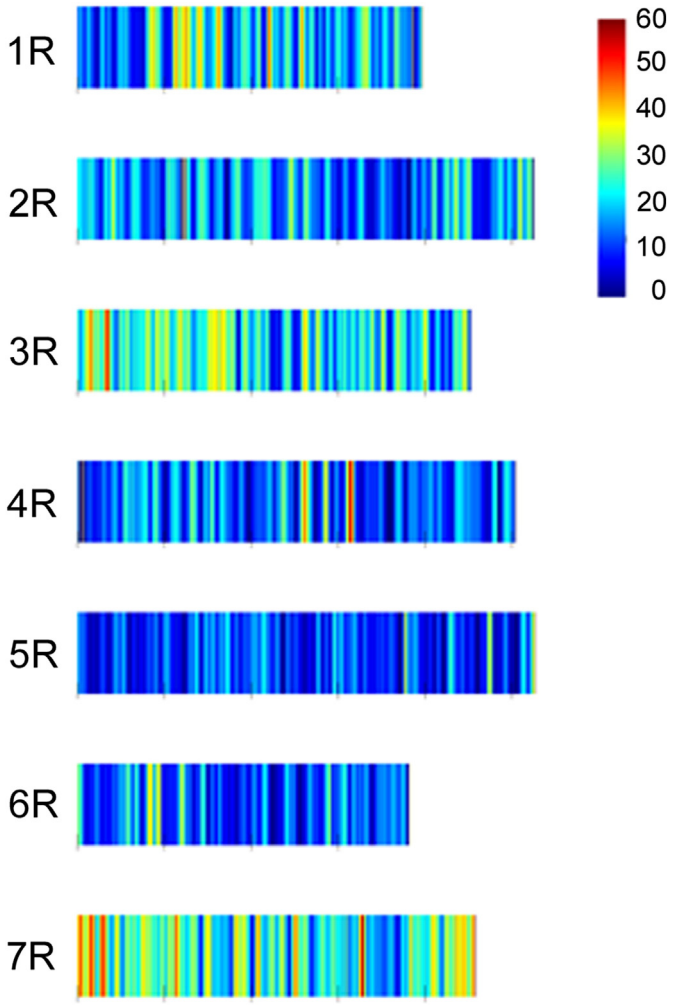


Fig. 1. Multi-A chromosomal origin of the rye B chromosome. Comparative analyses show the collinearity of the rye B with parts of the rye A chromosome complement. Gene related rye B sequences were *in silico* mapped onto the rye A chromosomes [15]. The heat map depicts the location of rye A genes tagged by sequences located on Bs. The homologous regions are colored yellow–red. The rye A chromosomes 3R and 7R are likely founder chromosomes of the ancient rye B chromosome. In addition, mainly chromosomes 1R and 2R contributed multiple small regions (Martis and Gundlach, unpublished).

theory of centromeric drive in female meiosis [36] accounts for the drive of Bs as proposed in [33,37] awaits experimental proof. The presence of fast-evolving repetitive sequences, along with reduced selective pressure on gene integrity, would predispose a nascent B to accumulate the further rapid structural modifications required to establish a drive mechanism. We expect Bs to be found primarily in taxa with elevated levels of chromosomal rearrangement and phylogenetic groups with unstable chromosome numbers. It is tempting to speculate that Bs originated as a by-product of A chromosome evolution [38,39].

3. Do B chromosomes carry active genes?

Regardless of their possible intra- or interspecies origin, A-derived sequences including genic sequences are potential components of any B. And as there is reduced selective pressure on genes in Bs, newly formed Bs should be richer in functional genic sequences than evolutionary older ones. The early studies addressing the presence of functional genes on Bs focused on those for which detection was simpler due to a high copy number (e.g. rRNA genes, reviewed in [40]). For instance, PCR-based analysis of the 40S precursor of rRNA suggested that the rDNA sequences on the large B of *Brachyscome dichromosomatica* are not transcribed [41], while in the grasshopper *Eyprepocnemis*

plorans B-derived rRNA transcripts were found. The different expression levels seen in different Bs suggest that structural or genetic features influence the generally weak activity of B-located rDNA [42,43]. Other high copy genes, like the B-located histone H3 and H4 genes of the migratory locust, are rather likely to be non-functional as higher sequence variation was found in them compared to their A-located counterparts [44].

Sequencing of flow-sorted [15] or microdissected [17,45] Bs has allowed to identify thousands of B-located genic sequences. The analysis of selected genic sequences showed that most of them have paralogous sequences on As (Fig. 2A) containing sequence polymorphisms. Some of the rye B genic sequences had lower similarity to their A-located counterparts reflecting their faster degeneration or earlier insertion in Bs [16,46].

In the cichlid *A. latifasciata* the majority of identified B-genes are fragmented. However, some genes on Bs were largely intact and the expression of the three of them could be confirmed [16]. In rye, at least parts of the B-genic sequences were found to be transcriptionally active. Some of them had the ability to modulate the expression of their counterparts on As in a tissue and genotype dependent manner [46]. A more advanced step of pseudogenization was found for the B-located rRNA genes of *E. plorans*. The Bs of this grasshopper contain additional 7000 to 15,000 rDNA units, but in contrast to the As, from 150 to over 1500 R2 non-LTR retrotransposons. The higher concentration of R2 in the inactive B-located rDNA clusters suggests that these chromosomes can act as a sink for R2 insertions thus further reducing the level of insertions on the As [47], if the total transposition rate is not increased.

A very young proto-B which likely evolved from heterochromatic A chromosome 4 within the last decade was found in a genetic stock of *Drosophila melanogaster*. Genetic assays and measuring the expression level of genes on chromosome 4 in individuals with and without Bs suggested that the B lacks the genes of the chromosome donor that were chosen for analysis. However, dependent on the genetic background, the B can act as an enhancer or suppressor of position-effect variegation [48]. In contrast, the B of *Drosophila albomicans* encodes transcriptionally active genes [49]. Transcriptome profiling of males with and without the B chromosome-like paternal sex ratio (PSR) chromosome performed in jewel wasp *Nasonia vitripennis* revealed nine PSR-specific transcripts. Four of them encoded for putative proteins and the rest are thought to produce long non-coding RNA [50].

Even Bs in mammals can be carriers of active genic sequences. Transcriptional activity was shown for a B-specific protein-coding sequence of the Siberian roe deer (*Capreolus pygargus*) in fibroblast culture [51]. This gene was seemingly derived from a duplicated A chromosome fragment. However, no *bona fide* B-derived protein has been demonstrated to date. As analysis of polyploids revealed a generally weak correlation between the transcript and protein levels of duplicated genes (reviewed in [52]), the chance of finding a B-specific protein might be rather low.

Besides gene coding sequences, retrotransposon-derived high-copy elements in maize and rye [26,53] and rye satellite repeat families including E3900 clustered on a B chromosome long arm are transcriptionally active [56]. The function and the mechanism of formation of these repeat-derived B transcripts are unknown. It has been hypothesized that these transcripts could have a function as scaffold RNA in the organization and regulation of Bs [26,54].

A comparative analysis of RNA-seq reads obtained from rye plants with and without Bs has been recently performed to assess the transcriptional activity of repetitive sequences. Transcripts derived from vegetative (root and leaf) and generative (anthers) tissues represented by 26–151 million RNA-seq reads were screened for their sequence similarity to a complete set of rye repetitive elements identified earlier [15]. Low levels of transcription were found for most repetitive elements, which represented a combined total of 3.1–4.9% of transcripts in analyzed

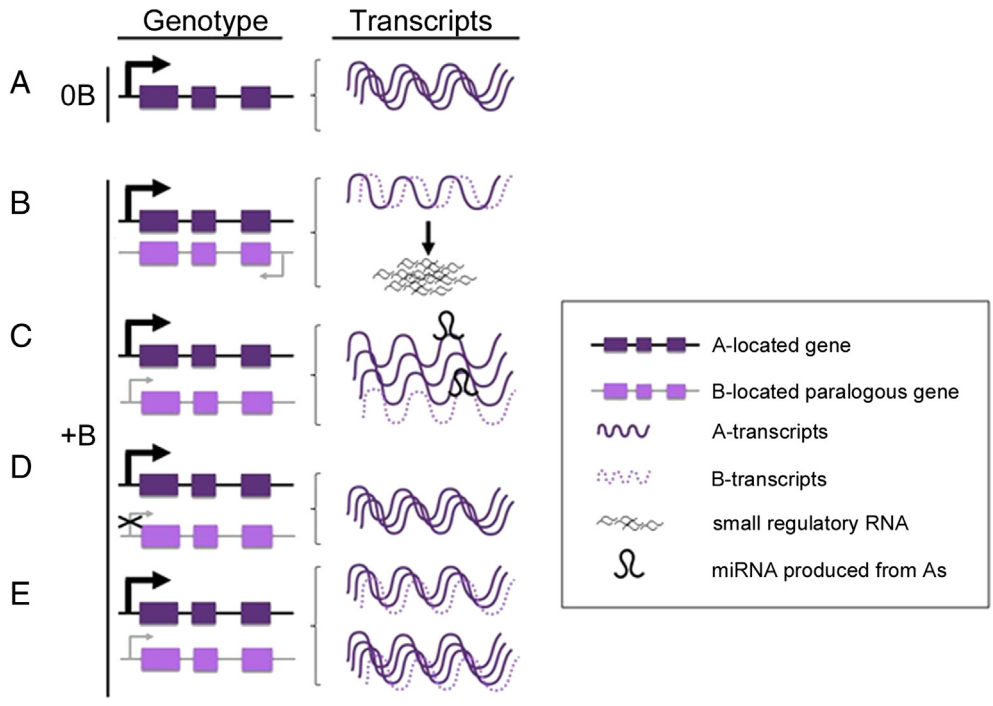


Fig. 2. Potential activity modulation of A chromosome-located genes by homologous sequences on B chromosomes. A) When the B chromosome is absent, a given gene on A has a defined level and pattern of expression. B) In the presence of a B, if the B-located gene is transcribed, transcripts could then serve as a substrate for RNA-directed RNA polymerase, which would make dsRNA, which is then processed to small RNAs. These small RNAs could cause mRNA degradation or epigenetic changes of A-located sequences. C) The B-transcripts could compete for regulatory factors responsible for regulation of A-located genes and could cause up- or down-regulation dependent on the mechanism of regulation by regulatory factors. D) In the presence of a B, if the corresponding B-located gene is inactive, the expression level and pattern of an A-located gene will not change. E) Finally, when both A- and B-located genes express from the same strand the expression level of that gene could remain unchanged compared to 0B due to dosage compensation or it causing upregulation due to accumulation of both A- and B-transcripts.

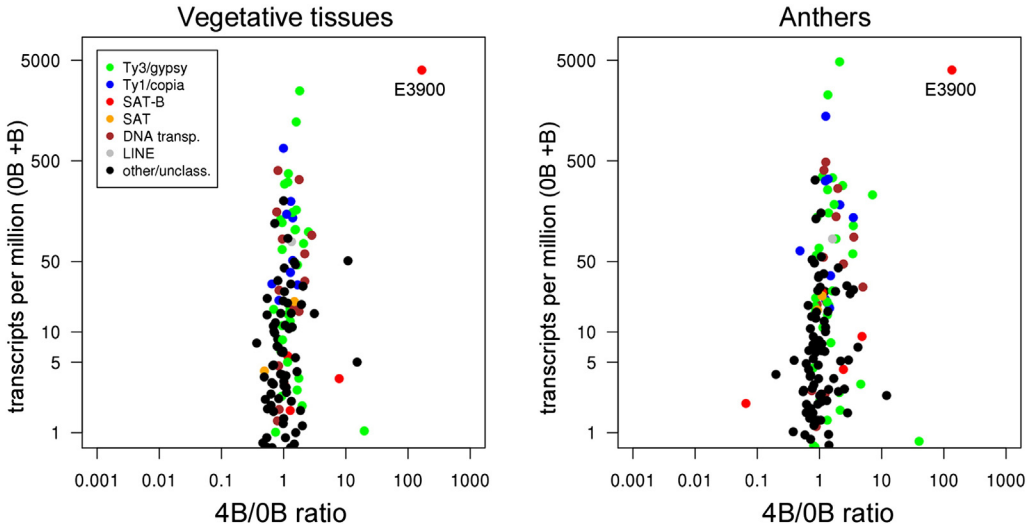


Fig. 3. Differential expression of repetitive sequences in rye plants with and without B chromosomes. Illumina reads (50 nt) obtained by sequencing of polyadenylated RNA from rye without (0B) and with (+4B) vegetative tissues (roots and leaves) or anthers were mapped based on their similarities to 200 most abundant rye repeats [15] represented by genomic reads identified using graph-based clustering analysis ([19], Suppl. material 1). These 200 repeat clusters represented 73% and 77% of A and B chromosome DNA, respectively. The plots show ratios of transcripts derived from individual repeats in plants containing (4B) *versus* lacking (0B) B chromosomes, and total abundance of these transcripts (normalized to 1 million analyzed reads). Dots on the plot represent repeat clusters and their colors correspond to repeat types (see the legend; satellites enriched/specific for B chromosomes (SAT-B) are distinguished from other satellite repeats (SAT)). Position of highly transcribed B-specific satellite E3900 is indicated.

tissues (excluding rRNAs). Repeat expression profiles were similar for plants with and without B chromosomes, with the exception of highly expressed B-specific satellite E3900, which was found in both vegetative tissues and anthers (Fig. 3, Macas et al., unpublished). This finding confirms previously reported data based on Northern blot experiments [54] . On the other hand, none of the other satellites enriched on or specific for B chromosomes [15,21] was found to be expressed at comparable levels. One explanation for this observation could be that in this analysis RNA preparations enriched for polyadenylated mRNAs were used to facilitate the detection of transcripts originating from rye B-located genic sequences.

4. Possible origins and fates of B-located genes

Besides being derived from ancestral founder A chromosomes, B-located genic sequences could also originate from horizontal gene transfer from As to established Bs. This could occur even from the genome of an intracellular parasite or symbiont. This scenario was approved by comparative sequencing of the cytoplasmic endosymbiont *Wolbachia pipientis* and its vector *Glossina morsitans morsitans*. Multiple insertions of *Wolbachia*-derived sequences, including *16S rRNA* and genes *fbpA*, and *wsp* were found in the B but also in the sex chromosomes of *G. morsitans morsitans*. Most of the transferred genes were found transcriptionally inactive. These data also suggest that the Bs in this species have originated from the Y chromosome, although the alternative possibility that the Y chromosome is derived from a B could not be ruled out [55]. Hitchhiking of genomic fragments from As by transposable elements could be a further possibility for the origin of B-located genes as demonstrated for non-collinear genes of Triticeae [56]. To address the exact mechanism underlying particular gene insertions, it will be necessary to analyze flanking regions of the respective B-located genic sequences. In analogy to homogeneously staining regions, chromosome aberrations that are characterized by the amplification of coding sequences, B-located coding and noncoding sequences seemingly are prone to amplification. Bs of the roe deer revealed regional amplification of A-derived sequences [51].

What is the likely evolutionary fate of B-located genic sequences? Because Bs are dispensable, it is expected that they are prone to accumulate mutations as they undergo pseudogenization. Depending on the species, the half-life of an active duplicated gene to become pseudogenized or be lost is estimated to be 2–7 million years [57]. Results from the human ENCODE project show that pseudogenes still can have function, for example by regulating their parental genes in different ways [58]. Pseudogenes can encode endo-siRNAs and regulate the expression of their parental gene [59]. Likely, as B-specific transcripts would be aberrant due to less selective pressure, they can serve as a substrate for RNA-directed RNA polymerase, which would make double strand RNA (dsRNA). The dsRNA is then processed to small regulatory RNA. This model is consistent with the cumulative effects of an increasing number of Bs (Fig. 2B). Alternatively, pseudogene transcripts can function as indirect post-transcriptional regulators. They might act as miRNA sponges. Due to high similarity between parental and pseudogene transcripts, both could compete for miRNA leading to degradation of parental gene transcripts (Fig. 2C) [60]. Further, it has been shown that some pseudogene transcripts translate and produce short peptides or truncated proteins with functional importance [59]. Pseudogene transcripts could also act as a source of siRNAs *via* pathways involving RNA-directed RNA polymerases. Comparative profiling of small RNAs from individuals with and without Bs could address the question whether B-derived siRNAs acting as regulators are indeed formed.

5. Does the chromatin structure differ between A and B chromosomes?

Although the chromatin structure is increasingly seen as playing essential roles in different aspects of chromosome function and gene regulation, little information is available on the chromatin composition of Bs. Based on classical cytological observations (*e.g.* Giemsa-banding) an early survey suggested that the Bs in about half of the plant species which carry them are heterochromatic [61]. However, in some species like the mouse *Apodemus flavicollis* [62], and the plant *Scilla vvedenskyi* [63] the Bs are even predominantly euchromatic.

DNA methylation and post-translational histone modifications are involved in chromatin assembly and maintenance [64–66]. The N-terminal tails of the nucleosomal core histones, extending from the nucleosome surface, are subjected to post-translational modifications such as acetylation, methylation, phosphorylation, ubiquitination, glycosylation, ADP-ribosylation, carbonylation and sumoylation. Acetylation of histones is mainly linked with transcriptional activation, DNA recombination and repair [67–70]. Phosphorylation correlates with transcription activation, apoptosis, DNA repair, chromosome condensation,

sister chromatid cohesion/segregation and gametogenesis [71]. The most well characterized histone modification to date is methylation. This modification occurs only on lysine (K) and arginine (R) residues of histones H3 and H4 and is performed by histone methyltransferases (HMTs) [72]. Each arginine can be mono- or dimethylated and each lysine can be either mono-, di-, or trimethylated [73]. Several studies have shown that modification of the histone H3 tail by methylation of lysine residues 9 and 27 negatively regulates transcription by mediating a compact chromatin structure. In contrast, euchromatin is marked by methylation of lysine residues 4 and 36 (reviewed by [72]). Whereas euchromatin-specific methylation of H3K4 is highly conserved among eukaryotes, heterochromatin indexing by methylation marks at H3K9, 27 and H4K20 is more variable (reviewed by [74]).

Although chromosome banding results suggest a similar eu- and heterochromatin composition in the B and A chromosomes of *B. dichromosomatica* (which carries large and micro-Bs), the use of specific antibodies against various lysine residues of histone H3 has shown that the Bs are characterized by a comparably low level of euchromatic histone marks. Micro-Bs which are heteropycnotic and tandem repeat-enriched revealed only traces of these histone modifications. No differences between As and Bs were found for the heterochromatic marks H3K9me1,2 and H3K27me1,2 indicating that Bs are not marked by an enriched level of heterochromatic histone marks either, but rather by a low level of euchromatin-associated histone modifications [75]. A comparable distribution of histone marks was also found for the Bs of *Crepis capillaris* [76] and for the heterochromatic Bs of *Puschkinia libanotica* [77].

In rye, the subterminal heterochromatic domain of the B is characterized by a unique combination of histone methylation marks [22,54]. Contrary to the heterochromatic regions of the A chromosomes, this domain is simultaneously marked by trimethylated histone H3K4 and trimethylated H3K27. In addition, this domain shows a dark Giemsa band at mitosis, but undergoes decondensation during interphase and reveals transcription of B-specific high copy repeat families. The distribution patterns along A and B chromosomes observed for the heterochromatin marks H3K9me1, 2 were mainly uniform. The terminal heterochromatic regions of As and Bs showed little H3K27me1 but were enriched in di- and trimethylated H3K27 [54].

Bs of maize display similar distributions of signals from H3K9me2, H3K27me2 and H3K27me1 compared with the As. However, the H3K27me2 signal on the B was less than those on the As. The H3K4me2 distribution pattern on the B matched the position of the cytologically visible euchromatin. Interestingly, only extremely faint H3K12ac signals were observed in the Bs [78]. A reduced level of histone H3 and H4 acetylation was also shown for the Bs of the grasshopper *E. plorans* [79] and of *B. dichromosomatica* [80], respectively.

Not only histones but also cytosine, which is one of the four essential base units of DNA nucleotides, can be extensively methylated. In both animals and plants, cytosine is primarily methylated at the CG dinucleotide position. However, in plants, methylation is not restricted to the CG sequence, but also occurs in CHG and less abundantly in CHH sequence contexts [81]. DNA methylation is involved in the silencing of transposable elements and genes [82]. The DNA repeats and mobile elements on Bs are also methylated, as has been reported for the Bd49 tandem repeat in *B. dichromosomatica* [83] and E3900 repeat in *Secale cereale* [84]. However, a higher DNA methylation level of Bs in comparison to As has not been reported for any plant species.

6. Many other open questions to be answered

The discovery of B-located genic sequences in a number of species raises many additional questions. As none of these Bs is completely sequenced and annotated, our knowledge about the proportion and structure of complete and truncated B-located genes is limited. Therefore, we can so far only speculate based on similar biological phenomena like *e.g.* sex chromosome evolution and polyploidization, where duplicated copies of genes coexisting with their parental genes are believed to play an important role in driving evolution [85].

Could any duplicated genes be located on B chromosomes? Dosage sensitive genes are less likely part of an evolving B, as duplication of a B chromosome-donor fragment containing such genes could result in a detrimental phenotype. In contrast, dosage insensitive genes like structural genes having no regulatory roles for transcription or translation [86] could be B-hosted and eventually undergo loss or pseudogenization. Duplicated genes are often associated with detrimental effects and being removed by natural selection. As most of the mutations are degenerative, it is more likely that a duplicated gene undergoes inactivation rather than to acquire a new function. Nevertheless, evidence exists for the beneficial role of duplicated genes especially in stressful conditions [87]. In some circumstances selective retention of duplicated genes could occur like, when their redundancy protects corresponding parental genes from immediate detrimental mutations, or when overdominance exists between their products. In this case duplicated genes could convert to new genes by achieving a beneficial mutation [88]. Therefore the possibility exists that B-located duplicated genes accelerate the evolution of their carriers.

How are B-located genes regulated? Transcripts from rye B-genic fragments undergo splicing and show tissue-dependent expression patterns [46]. The pattern and level of their expression could differ from their paralogues on As if mutations changed their regulatory elements (Fig. 2A and D). On the other hand the regulation of B-located genes and their effects on A-located genes could vary between different species (Fig. 2E).

Expression changes caused by aneuploidy have often genome-wide impacts and the spectrum of expression changes is often complex [89]. In maize, compensation for the majority of genes on aneuploid chromosomes occurs at the RNA level [90]. For plants evidence exists that *de novo* formed genes show high epigenetic variations among different individuals which was reflected in their expression patters. It is proposed that newly formed genes are especially prone at the initial stage of their formation to epigenetic variation [91]. It is unknown whether the same is true for B-located genes.

Could a B-located genic sequence with the ability to modulate the activity of a corresponding A-located gene cause evolutionary effects at the long term? If the product of the B-located gene would become beneficial, positive selection would act to keep the respective B present in the population. On the other hand, this could release the pressure on a B-specific drive mechanism, meaning a former selfish B chromosome would convert into a beneficial genome component. In such a way, the Y chromosome of *Drosophila* may have evolved from a supernumerary chromosome with the characteristics of a B chromosome [92]. Accumulating evidence exists that in addition to genetic variation, epigenetic variations exist among individuals of a population [93]. These epigenetic variations are triggered by genetic variations as well as different environmental stimuli. It would be worth testing whether Bs affect the epigenetic status of As by comparing epigenetic modifications and expression levels of parental A genes in the presence and absence of Bs for several generations.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.bbagrm.2014.11.007.

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